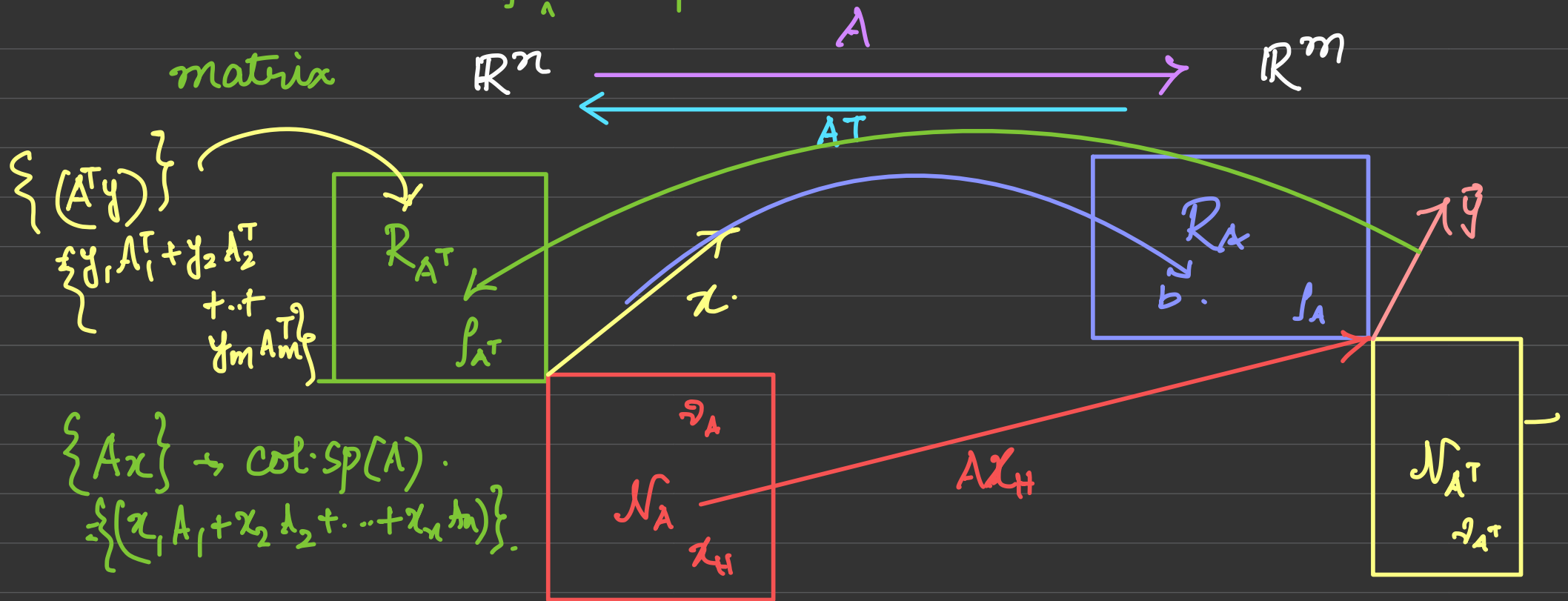


four fundamental. Orientation of subspaces associated with a matrix

04.01.2026.



For $x \in \mathcal{N}_A$, $Ax = 0^m$

$y \cdot Ax = 0$ for $x \in \mathcal{N}_A$, for every $y \in \mathbb{R}^m$ (scalar)

$\Rightarrow (Ax)^T y = 0$ for $x \in \mathcal{N}_A$, for every $y \in \mathbb{R}^m$

$\Rightarrow x^T A^T y = 0$ " " " " " " "

$= x^T (A^T y) = 0$ " " " " " "

$\Rightarrow x^T (A^T y) = 0$ for $x \in N_A$, for every $y \in \mathbb{R}^m$
 for every $y \in \mathbb{R}^m$ $A^T y$ is an element of $\text{col.sp}(A^T)$.

$$\Rightarrow x^T (A^T y) = 0 \quad x \in N_A.$$

$$= x \cdot (A^T y) = 0 \quad \underbrace{x \in N_A}, \quad \underbrace{A^T y \in R_{A^T}}.$$

$$\Rightarrow N_A \perp R_{A^T}.$$

$\Rightarrow N_A$ orthogonal compl. of R_{A^T}

$$\Rightarrow N_A = R_{A^T}^\perp \quad R_{A^T} = W, \quad W^\perp = R_{A^T}^\perp = N_A.$$

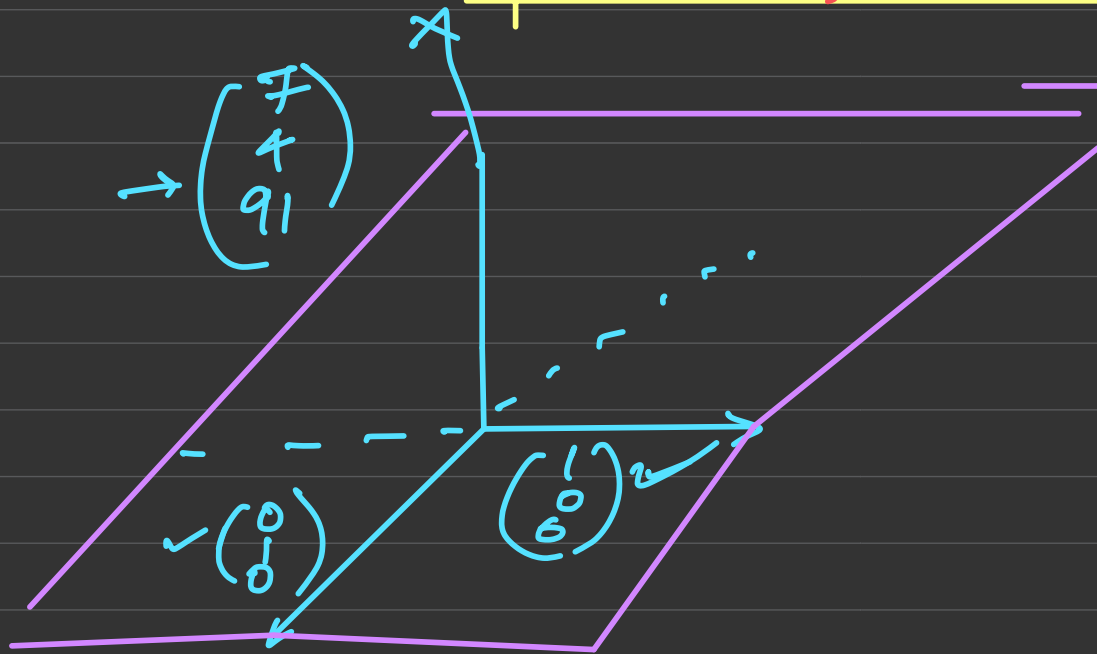
$$R_{A^T} = N_A^\perp$$

$$N_A = W, \quad W^\perp = N_A^\perp = R_{A^T}.$$

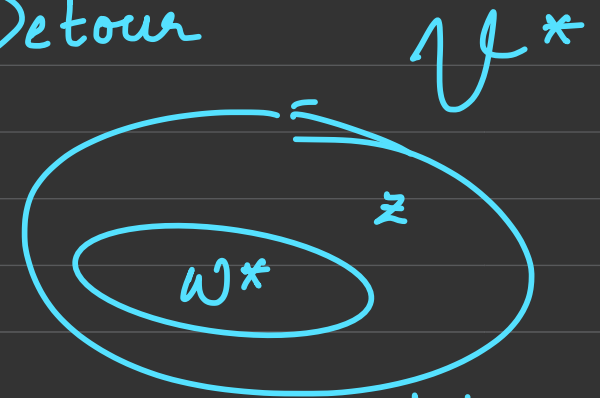
Similarly we can prove $N_{A^T} \perp R_A$

$$\Rightarrow N_{A^T}^\perp = R_A; \quad R_A^\perp = N_{A^T}.$$

$$V = \mathbb{R}^k$$



Detour



$u, v \in W^*$ & l.i.

$z \in V^*$, but not in W^* .

Claim: u, v, z are l.i.

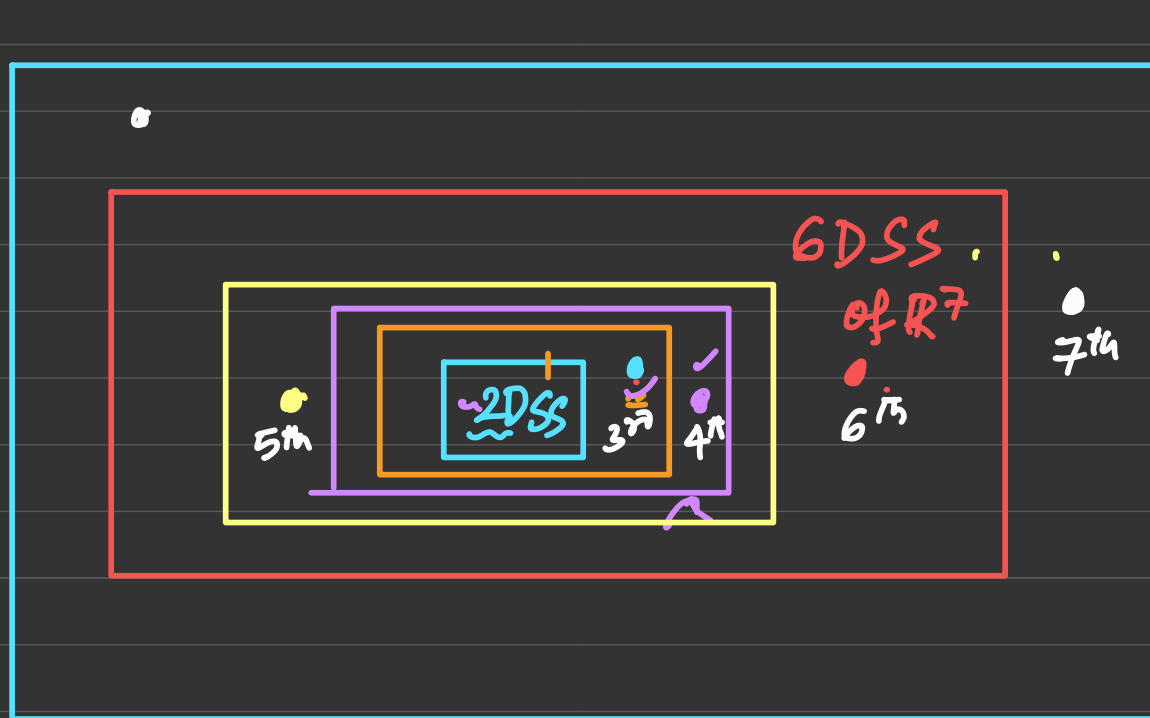
Proof: $\alpha u + \beta v + \gamma z = \vec{0}$

$$\Rightarrow \gamma = 0.$$

If $\gamma \neq 0$, $\alpha \vec{u} + \beta \vec{v} + \gamma \vec{z} = \vec{0}$

$$\Rightarrow \gamma \vec{z} = -\alpha \vec{u} - \beta \vec{v}$$

$$= \vec{z} = -\frac{\alpha}{\gamma} \vec{u} - \frac{\beta}{\gamma} \vec{v}.$$



$$\mathbb{R}^7.$$

Any set of
 $(n+1)$ vectors
 in $\mathbb{R}^n$ is
 ALWAYS a
 linearly dep set.

Given a vector space V of dim K , and a subspace W of dim d , of V , one can extend the basis for W to the basis for V by looking at vectors outside of W along with the basis for W .

$\mathcal{U} \cdot \dim K.$

	$\vec{x} \in W^\perp \dim?$
W $\phi_1, \phi_2 \dots \phi_d$ $\dim = d.$	

Let $\phi_1, \phi_2 \dots \phi_d$ be an orthonormal basis for W .

$$\phi_i = \begin{pmatrix} \phi_{i1} \\ \phi_{i2} \\ \vdots \\ \phi_{ik} \end{pmatrix} \text{ for } i = 1 \dots d$$

Let us extend this ONB to an orthonormal basis for $\mathcal{U}$

i.e.,

$$\phi_1, \phi_2 \dots, \phi_d, \chi_1, \chi_2 \dots \chi_{k-d}$$

Let $\vec{x} \in W^\perp \Rightarrow \vec{x} \cdot \vec{w} = 0$ for every vector $w \in W$.

$$\vec{w} \in W = (w \cdot \phi_1) \vec{\phi}_1 + (w \cdot \phi_2) \vec{\phi}_2 + \dots + (w \cdot \phi_d) \vec{\phi}_d \quad \text{Basis exp'n for } W.$$

$$\Rightarrow \text{If } x \in W^\perp, \quad x \cdot w = 0 \text{ for } w \in W$$

$$\text{then } (x \cdot \phi_i) = 0 \text{ for } i = 1 \dots d.$$

$$\text{Conversely if } (x \cdot \phi_j) = 0 \Rightarrow x \cdot \sum_{j=1}^d \alpha_j \phi_j = 0 \text{ for } j=1 \dots d. \quad \alpha_j \in \mathbb{R}.$$

$$\Rightarrow x \cdot w = 0 \text{ for all } w \in W.$$

$$\Rightarrow x \in W^\perp$$

Recall $x_1, x_2 \dots x_{k-d}$ is a Orthonormal Set & this

Set is also orthogonal to $\phi_1, \phi_2 \dots \phi_d$.

$$\therefore x_1, x_2 \dots x_{k-d} \in W^\perp$$

Q: Does $x_1, x_2, \dots, x_{k-d}$ span $W^\perp$?

Consider a vector $x \in W^\perp$

$\vec{x}$ has an ONB expansion given by

$$\vec{x} = \underbrace{(x \cdot \vec{\phi}_1)}_0 \vec{\phi}_1 + \underbrace{(x \cdot \vec{\phi}_2)}_0 \vec{\phi}_2 + \dots + \underbrace{(x \cdot \vec{\phi}_d)}_0 \vec{\phi}_d + (x \cdot x_1) x_1 + \dots + (x \cdot x_{k-d}) x_{k-d}.$$

$$\Rightarrow \vec{x} = (x \cdot x_1) \check{x}_1 + \dots + (x \cdot x_{k-d}) \check{x}_{k-d}$$

Recall $x_1, \dots, x_{k-d}$ are orthonormal & hence l.i.

$$\Rightarrow \vec{x} \in \underbrace{\text{Span}(x_1, x_2, \dots, x_{k-d})}_{\text{vector space}}. \quad \begin{array}{l} x \in W^\perp \rightarrow \text{VS.} \\ x \in \text{Span}(x_1, \dots, x_{k-d}) \end{array}$$

$\Rightarrow W^\perp$ is spanned by $x_1 \dots x_{k-d}$ & hence $x_1 \dots x_{k-d}$ is a basis for $W^\perp$
 $\Rightarrow W^\perp$ is a VSS of $\dim(k-d)$.

$$\therefore \dim(W) + \dim(W^\perp) = d + k - d = k = \dim(V).$$

As a consequence $\dim(\text{col sp}(A^T)) + \dim(N_A) = \dim(\mathbb{R}^n)$

$$= r_{A^T} + n_A = n.$$

||| only

$$r_A + n_{A^T} = m.$$